

Food Supplementation in Patients Hospitalized for Heart Failure A Randomized Clinical Trial

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 Supplemental content

IMPORTANCE Low-quality dietary intake is associated with adverse heart failure (HF) outcomes, yet evidence evaluating food-as-medicine interventions that supply high-quality dietary content is limited.

OBJECTIVE To determine the feasibility of providing food supplementation with medically tailored meals or fresh produce and explore the association of food supplementation vs usual care with clinical outcomes in patients recently hospitalized with HF and whether it differs by conditioning food supplementation to health care engagement.

DESIGN, SETTING, AND PARTICIPANTS This open-label factorial randomized clinical trial was conducted between April 2024 and October 2025 at 2 hospitals in Dallas, Texas. The study included patients who were hospitalized for HF and enrolled within 14 days of discharge, excluding those with prior heart transplant, a left ventricular assist device, or inotropic support at discharge; current enrollment in meal delivery programs; and inability to receive home deliveries. Participants were followed up with for 12 weeks. Of 150 participants enrolled, 2 were withdrawn due to clinical deterioration, 1 died, and 6 were lost to follow-up; all were included in the intention-to-treat analysis.

INTERVENTIONS Participants were randomized 1:1:1 to medically tailored meals, fresh produce, or usual care. Those receiving food supplementation underwent secondary 1:1 randomization to conditional (linked to clinic attendance and medication fills) vs unconditional delivery.

MAIN OUTCOMES AND MEASURES Implementation outcomes included delivery completion, adherence, and acceptability and exploratory clinical outcomes. The primary clinical outcome was defined as readmission for HF or emergency department (ED) visits for HF over 90-day follow-up. Secondary clinical outcomes included a win-ratio-based hierarchical composite (all-cause death, total HF hospitalizations or ED visits, and ≥ 10 -point Kansas City Cardiomyopathy Questionnaire-Clinical Summary Score [KCCQ-CSS] improvement).

RESULTS Among 150 participants (median [IQR] age, 59.5 [52.0-66.0] years; 91 [60.7%] male), the baseline median (IQR) left ventricular ejection fraction was 35% (25.0-54.0), the baseline median (IQR) KCCQ-CSS was 56.6 (36.8-72.9), and 79 (52.7%) had food insecurity. Food delivery completion was 93.6% with a mean (SD) reported consumption adherence of 4.7 (2.4) days per week (medically tailored meals) and 5.5 (2.3) days per week (fresh produce), with high retention (96.0%). Fresh produce demonstrated superior acceptability compared with medically tailored meals (Net Promoter Score: 8.6 vs 7.3; $P = .02$). There was no significant difference in the primary clinical outcome (HF readmission or ED visit) between food supplementation (23 events among 100 participants) vs usual care (9 events among 50 participants) (adjusted rate ratio, 1.09; 95% CI, 0.49-2.43; $P = .83$). The hierarchical composite favored food supplementation vs usual care (win ratio, 1.21; 95% CI, 1.14-1.29; $P < .001$). Among participants receiving food supplementation, conditional delivery (vs unconditional) was not associated with risk of HF events.

CONCLUSIONS AND RELEVANCE In this randomized clinical trial, 90-day food supplementation was feasible and well accepted in individuals with recent HF. Food supplementation was not associated with an improvement in the primary clinical outcome and there was no difference when conditioning food supplementation on health care engagement.

TRIAL REGISTRATION ClinicalTrials.gov Identifier: [NCT06115369](https://clinicaltrials.gov/ct2/show/study/NCT06115369)

JAMA Cardiol. doi:[10.1001/jamacardio.2026.0435](https://doi.org/10.1001/jamacardio.2026.0435)
Published online April 8, 2026.

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Despite advances in pharmacologic therapy and disease management, outcomes following hospitalization for acute decompensated heart failure (HF) remain poor.¹⁻³ Multiple factors contribute to postdischarge vulnerability, including inadequate HF medication utilization, suboptimal follow-up, dietary challenges, limitations in physical activity, and complex self-care requirements.⁴⁻⁶ Among these, food insecurity and poor diet quality affect up to 50% of patients with HF in the postdischarge period and are strongly associated with malnutrition, functional decline, readmission, and mortality.⁷⁻¹² Thus, interventions to improve nutritional status represent a potential strategy to improve postdischarge HF outcomes.

Food-as-medicine interventions integrate food and nutrition support into health care delivery.⁸ Medically tailored meals provide fully prepared, nutritious, ready-to-eat meals tailored to specific medical conditions; medically tailored groceries deliver unprepared healthy foods for preparation at home; and produce prescriptions increase access to fruits and vegetables through vouchers or direct provision. Prior studies of food-based interventions in outpatients with chronic cardiometabolic conditions such as diabetes and hypertension have demonstrated heterogeneous results.¹³ Evidence is more limited among patients following HF hospitalization, with few trials investigating food supplementation or nutritional interventions.¹⁴⁻¹⁶ Furthermore, the comparative effectiveness of different food-as-medicine approaches remains unknown, and whether food supplementation can be leveraged to incentivize postdischarge health care engagement has not been evaluated.

We conducted a pilot factorial randomized clinical trial to determine the feasibility and acceptability of 2 different forms of food supplementation, medically tailored meals or fresh produce, vs usual care in patients who were recently hospitalized with HF. We also explored the association of food supplementation with clinical outcomes and quality of life. Finally, we evaluated whether conditioning the food intervention to participants' clinic attendance and medication pharmacy fills was associated with better outcomes than unconditional delivery.

Methods

Clinical Trial Design

The Examining the Impact of Medically Tailored Meals vs Produce Supplements Delivered Conditionally vs Not on Health Engagement and Health Outcomes Among Patients With Heart Failure (FOOD-HF) trial was conducted between April 2024 and October 2025 across 2 centers: Parkland Hospital (urban county safety-net hospital) and Clements University Hospital (academic hospital), in Dallas, Texas. The randomized clinical trial was open label and used a 3 × 2 factorial design. The first randomization assigned participants in a 1:1:1 ratio to medically tailored meals, fresh produce boxes, or usual care with dietary counseling only. Food supplementation participants underwent secondary randomization 1:1 to conditional vs unconditional delivery. Under conditional delivery, continued food provision was linked to health care engagement, including attendance at clinic visits and medication adherence.

Key Points

Question Is food supplementation with delivery of medically tailored meals or fresh produce feasible and does it affect clinical outcomes compared with usual care in patients recently hospitalized with heart failure (HF)?

Findings In this randomized clinical trial of 150 participants, 90-day food supplementation demonstrated high feasibility and acceptability, with fresh produce superior to medically tailored meals. Food supplementation was not associated with reduced hospitalizations or emergency department visits for HF with or without delivery conditional on health care engagement.

Meaning The results of this study indicate that food supplementation was feasible and well accepted but did not demonstrate differences in clinical outcomes.

The trial was developed as part of the American Heart Association Health Care by Food Initiative.⁸ Protocol development occurred collaboratively with input from the initiative's Research Planning Group and Cooperative Studies Group, which included multidisciplinary expert review in biostatistics, behavioral science, patient-centered design, and implementation science (eMethods in [Supplement 2](#)). The protocol was approved by the University of Texas Southwestern Medical Center institutional review board and registered at ClinicalTrials.gov (NCT06115369) and is available in [Supplement 1](#). All participants provided written informed consent.

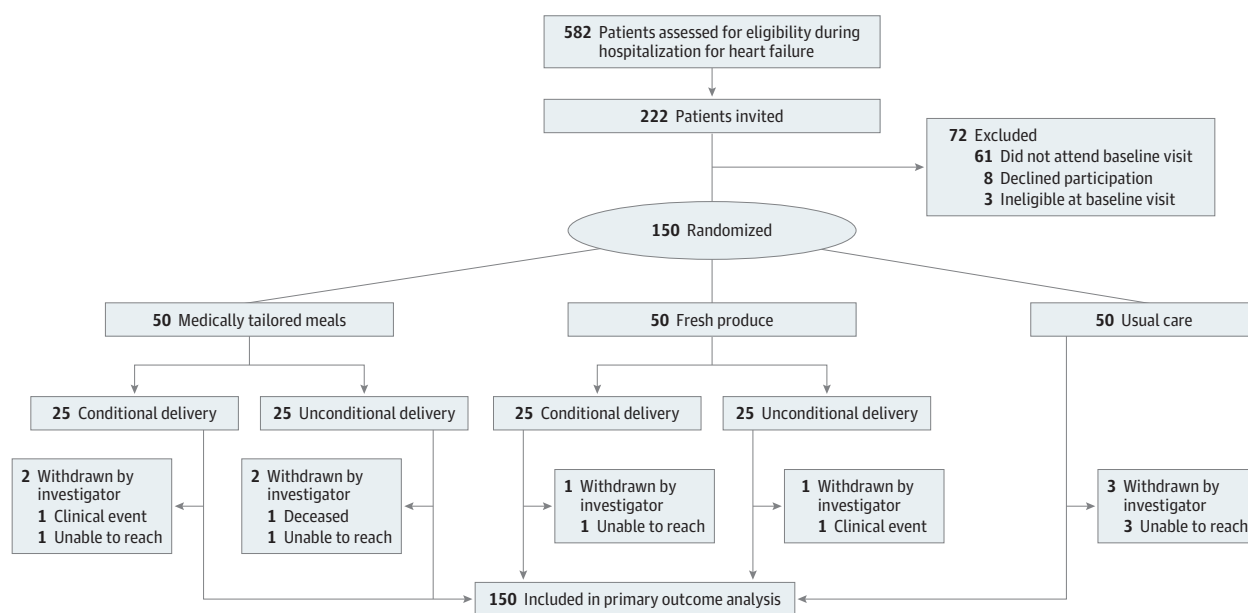
Study Population

Eligible patients were identified during hospitalization for acute decompensated HF, approached prior to discharge, and invited to outpatient study visit within 14 days of discharge to obtain informed consent and complete baseline assessments. Key exclusion criteria included prior heart transplant, left ventricular assist device or inotropic support at discharge, current enrollment in meal delivery programs, and inability to receive home deliveries. Complete inclusion and exclusion criteria are provided in eTable 1 in [Supplement 2](#). Race and ethnicity were collected by participant self-report using fixed categories (American Indian or Alaska Native, Asian, Black, Native Hawaiian or Other Pacific Islander, White, or other, which included participants who did not self-identify with the prespecified race categories, for race; Hispanic or non-Hispanic for ethnicity). These data were collected because food and nutrition insecurity disproportionately affect racial and ethnic minority populations with HF.¹⁰

Food Supplementation Interventions and Usual Care

All food interventions were delivered through Season Health, a commercial food-as-medicine vendor integrating registered dietitian-led medical nutrition therapy with meal and grocery delivery.¹⁷ Participants in both food intervention arms received virtual dietitian consultations at randomization, 30 days, and 60 days for disease-specific dietary education and personalized food selection. Both

Figure 1. CONSORT Flow Diagram



Flow diagram showing participant screening, enrollment, randomization, and follow-up in the FOOD-HF trial. A total of 150 patients hospitalized for acute decompensated heart failure were randomized 1:1 to medically tailored meals (n = 50), fresh produce boxes (n = 50), or usual care with dietary counseling

(n = 50). Participants randomized to food interventions underwent secondary 1:1 randomization to conditional (n = 50) vs unconditional (n = 50) delivery. All 150 randomized participants were included in the intention-to-treat analysis.

arms had access to a self-directed, 12-week HF education curriculum covering nutrition, sodium and fluid management, medication adherence, and physical activity (eTable 2 in [Supplement 2](#)).

The food interventions followed a low-sodium, vegetable-forward dietary pattern influenced by Dietary Approaches to Stop Hypertension (DASH) principles but adapted for HF, prioritizing sodium restriction (<2000 mg daily) while incorporating increased vegetables, whole grains, and lean proteins. Personalization included modifications for comorbid diabetes (lower carbohydrate emphasis, glycemic index considerations) or chronic kidney disease (avoidance of high-potassium foods). Cultural tailoring incorporated diverse meal and produce options reflecting regional cuisines, including Latin-style ingredients and recipes.

Participants assigned to medically tailored meals received 14 fully prepared, refrigerated meals weekly (2 meals per day) for 90 days, averaging approximately 650 mg sodium per meal (target <2000 mg daily). Sample meal options and average nutritional profiles are provided in eTable 3 in [Supplement 2](#). Participants assigned to fresh produce boxes received weekly deliveries of fresh fruits, vegetables, whole grains, dairy products, lean protein, olive oil, spices, and pantry staples, plus recipes for meal preparation. Produce box sample contents are detailed in eTable 4 in [Supplement 2](#). Deliveries occurred weekly on participant-selected days. Participants in usual care received dietary counseling about sodium restriction, fluid management, and heart-healthy eating patterns at the time of hospital discharge using the American Heart Association Discharge Packet for HF.

Conditional vs Unconditional Food Supplement Delivery and Follow-Up

Participants receiving food interventions were randomized 1:1 to unconditional delivery (all weekly shipments regardless of health care engagement) or conditional delivery. Continued provision was associated with attendance at appointments and medication fills (eMethods in [Supplement 2](#)). Details about baseline and follow-up visit assessments, which included in-person visits at baseline and end of study (week 12) and telephone call follow-up at weeks 4 and 8, are provided in the eMethods in [Supplement 2](#).

Study Outcomes

Implementation outcomes were selected based on the Reach, Effectiveness, Adoption, Implementation, Maintenance (RE-AIM) framework. Prespecified implementation outcomes were feasibility, acceptability, and appropriateness of the food interventions. Feasibility included successful delivery rates and 7-day recall of meals or produce consumed from the program. The acceptability, appropriateness, and feasibility of intervention measure were assessed at 90 days using validated 4-item scales for each construct (range, 1-5; higher scores indicate greater agreement).¹⁸ Participant experience was measured using the Net Promoter Score (range 0-10), with scores 9 or higher classified as promotional, 7-8 as passive, and 6 or lower as detractive.¹⁹

The primary clinical outcome was a composite of total readmissions for HF or emergency department (ED) visits for HF within 90 days of randomization. Two physician investigators (N.K. and S.R.) independently adjudicated all events to

Table 1. Baseline Characteristics Across Food Supplementation Randomization Groups

	No. (%)				
Characteristic	Overall (N = 150)	Control (n = 50)	Fresh produce (n = 50)	Medically tailored meals (n = 50)	P value ^a
Demographic characteristics					
Age, median (IQR), y	59.5 (52.0-66.0)	60.5 (53.0-66.0)	58.5 (48.0-65.0)	59.0 (53.0-68.0)	.64
Sex					
Female	59 (39.3)	20 (40.0)	23 (46.0)	16 (32.0)	.36
Male	91 (60.7)	30 (60.0)	27 (54.0)	32 (68.0)	
Race ^b					
Asian	2 (1.3)	1 (2.0)	0 (0.0)	1 (2.0)	.86
Black	63 (42.0)	23 (46.0)	18 (36.0)	22 (44.0)	
White	80 (53.3)	25 (50.0)	30 (60.0)	25 (50.0)	
Other	5 (3.3)	1 (2.0)	2 (4.0)	2 (4.0)	
Hispanic ethnicity ^b	49 (32.7)	13 (26.0)	19 (38.0)	17 (34.0)	.43
Uninsured or county-sponsored coverage	57 (38.0)	16 (32.0)	19 (38.0)	22 (44.0)	.47
Married or living with partner	62 (41.3)	19 (38.0)	22 (44.0)	21 (42.0)	.82
Some high school education or less	16 (10.7)	8 (16.0)	5 (10.0)	3 (6.0)	.26
Unemployed/retired	99 (66.0)	37 (74.0)	29 (58.0)	33 (66.0)	.24
Self-reported financial strain	75 (50.0)	25 (50.0)	23 (46.0)	27 (54.0)	.73
Enrolled from county hospital	93 (62.0)	35 (70.0)	29 (58.0)	29 (58.0)	.36
Clinical characteristics					
Baseline systolic blood pressure, median (IQR), mm Hg	117 (104-136)	112 (104-124)	121 (105-145)	118 (102-136)	.41
Baseline BMI	31.3 (27.5-37.6)	32.8 (27.4-40.5)	32.1 (29.9-36.8)	29.6 (26.5-35.3)	.05
Screening LVEF, median (IQR), %	35.0 (25.0-54.0)	39.5 (24.0-55.0)	35.5 (25.0-56.0)	29.5 (25.0-44.0)	.54
EF ≥50%	43 (28.7)	16 (32.0)	18 (36.0)	9 (18.0)	.11
Baseline NT-proBNP, median (IQR), pg/mL	1197.0 (436.0-2673.0)	761.0 (319.0-2518.0)	1217.5 (479.0-2407.0)	1559.0 (615.0-3706.0)	.04
Screening eGFR, median (IQR), mL/min/1.73 m ²	57.0 (42.0-77.0)	64.5 (52.0-79.0)	54.5 (38.0-78.0)	54.0 (40.0-75.0)	.14
CKD stage					
II	40 (33.1)	16 (41.0)	11 (27.5)	13 (31.0)	.60
III	65 (53.7)	19 (48.7)	23 (57.5)	23 (54.8)	
IV	14 (11.6)	4 (10.3)	6 (15.0)	4 (9.5)	
V	2 (1.7)	0 (0.0)	0 (0.0)	2 (4.8)	
Hypertension	142 (94.7)	48 (96.0)	46 (92.0)	48 (96.0)	.73
Diabetes	81 (54.0)	30 (60.0)	20 (40.0)	31 (62.0)	.05
Hyperlipidemia	122 (81.3)	45 (90.0)	39 (78.0)	38 (76.0)	.15
CVA	20 (13.3)	11 (22.0)	3 (6.0)	6 (12.0)	.06
Coronary artery disease	59 (39.3)	22 (44.0)	16 (32.0)	21 (42.0)	.42
COPD	46 (30.7)	18 (36.0)	15 (30.0)	13 (26.0)	.55
NYHA functional class					
I	8 (5.3)	2 (4.0)	5 (10.0)	1 (2.0)	.29
II	67 (44.7)	22 (44.0)	25 (50.0)	20 (40.0)	
III	75 (50.0)	26 (52.0)	20 (40.0)	29 (58.0)	
HF medical therapy					
ACEI/ARB/ARNI	121 (80.7)	44 (88.0)	36 (72.0)	41 (82.0)	.12
β-Blocker ^c	123 (82.0)	39 (78.0)	41 (82.0)	43 (86.0)	.58
MRA	95 (63.3)	32 (64.0)	32 (64.0)	31 (62.0)	.97
SGLT-2 inhibitor	99 (66.0)	30 (60.0)	30 (60.0)	39 (78.0)	.09
Loop diuretic	143 (95.3)	48 (96.0)	47 (94.0)	48 (96.0)	>.99

(continued)

confirm HF as the primary reason for presentation, with disagreements resolved by consensus. Additional prespecified exploratory clinical outcomes included a win-ratio-based, hierarchical composite of (1) all-cause death, (2) total HF hospitalizations or ED visits for HF, and (3) clinically meaningful improvement in health status (≥10-point increase in Kansas City

Cardiomyopathy Questionnaire-Clinical Summary Score [KCCQ-CSS]).²⁰ Win-ratio methodology compares participants between groups through the outcome hierarchy. A participant achieves a win if they have a better outcome at the first level where outcomes differ, and the win ratio is calculated as total wins divided by total losses, with values greater

Table 1. Baseline Characteristics Across Food Supplementation Randomization Groups (continued)

	No. (%)				
Characteristic	Overall (N = 150)	Control (n = 50)	Fresh produce (n = 50)	Medically tailored meals (n = 50)	P value ^a
Index hospitalization characteristics					
Discharge to randomization, median (IQR), d	6.0 (4.0-8.0)	6.5 (3.0-10.0)	6.0 (5.0-8.0)	6.0 (4.0-7.0)	.54
Length of index hospitalization, median (IQR), d	4.0 (3.0-5.0)	4.0 (2.0-6.0)	4.0 (3.0-6.0)	4.0 (3.0-5.0)	.34
Patient-reported outcomes					
Baseline KCCQ-CSS, median (IQR)	56.6 (36.8-72.9)	57.7 (28.8-77.6)	55.9 (38.5-74.0)	57.9 (40.6-72.4)	.94
Self-reported adherence (MMAS-8)					
High adherence	47 (31.3)	14 (28.0)	15 (30.0)	18 (36.0)	.76
Medium adherence	57 (38.0)	18 (36.0)	19 (38.0)	20 (40.0)	
Low adherence	46 (30.7)	18 (36.0)	16 (32.0)	12 (24.0)	
Nutrition security					
Nutrition insecure	82 (55.4)	29 (60.4)	24 (48.0)	29 (58.0)	.42
Nutrition secure	66 (44.6)	19 (39.6)	26 (52.0)	21 (42.0)	
Food security					
Food insecurity	79 (52.7)	27 (54.0)	24 (48.0)	28 (56.0)	.71
Food security	71 (47.3)	23 (46.0)	26 (52.0)	22 (44.0)	

Abbreviations: ACEI, angiotensin-converting enzyme inhibitor; ARB, angiotensin receptor blocker; ARNI, angiotensin receptor-neprilysin inhibitor; BMI, body mass index (calculated as weight in kilograms divided by height in meters squared); COPD, chronic obstructive pulmonary disease; CVA, cerebrovascular accident; EF, ejection fraction; eGFR, estimated glomerular filtration rate; HF, heart failure; KCCQ-CSS, Kansas City Cardiomyopathy Questionnaire Clinical Summary Score; LVEF, left ventricular ejection fraction; MMAS-8, 8-item Morisky Medication Adherence Scale; NT-proBNP, N-terminal pro-B-type natriuretic peptide; NYHA, New York Heart Association; SGLT-2, sodium-glucose cotransporter-2.

^a P values represent comparisons across the three randomization groups using Kruskal-Wallis rank sum test for continuous variables and Pearson χ^2 test for

categorical variables.

^b Race and ethnicity were collected by participant self-report using fixed categories (American Indian or Alaska Native, Asian, Black, Native Hawaiian and Other Pacific Islander, White, or other, which included individuals who did not self-identify with the fixed categories, for race; Hispanic or non-Hispanic for ethnicity). These data were collected because food and nutrition insecurity disproportionately affect racial and ethnic minority populations with heart failure.¹⁰

^c Among patients with EF <50% (n = 107: control n = 34, fresh produce n = 41, medically tailored meals n = 32).

than 1 favoring the intervention. Deaths were ascertained through health record review or next of kin. Additional secondary outcomes included change in KCCQ-CSS (range 0-100; higher scores indicate better health status) from baseline to 90 days.²¹ Healthcare process measures included percent attendance at outpatient appointments and percent of medication fills through 90-day follow-up.

Statistical Analysis

The FOOD-HF pilot trial enrolled 150 participants based on feasibility considerations balancing scientific objectives with available resources. A formal power calculation was not performed owing to the pilot nature of the study. The clinical outcome analysis followed intention-to-treat principles, including all randomized participants censored at the time of last known follow-up, time of death, or 90 days.

Baseline characteristics were summarized by treatment group using medians (IQRs) for continuous variables and frequencies (%) for categorical variables and compared across treatment groups using Kruskal-Wallis tests for continuous variables and χ^2 for categorical variables. Feasibility measures (delivery success rates, food consumption frequency), Acceptability of Intervention Measure, Intervention Appropriateness Measure, and Feasibility of Intervention Measure and Net

Promoter Score were summarized by food supplementation group and compared using Wilcoxon rank sum tests for continuous variables and χ^2 tests for categorical variables.

HF hospitalizations and ED visits for HF were analyzed using Poisson regression with an offset term to estimate rate ratios and 95% CIs. The factorial included 2 comparison sets. For food supplementation effects, comparisons included pooled food interventions vs usual care, medically tailored meals vs usual care, and fresh produce boxes vs usual care. For conditionality effects, comparisons included conditional vs unconditional delivery, conditional delivery vs usual care, and unconditional delivery vs usual care. Adjusted models included study site, age, sex, left ventricular ejection fraction, and baseline estimated glomerular filtration rate as covariates.

The win-ratio hierarchical composite compared pooled food supplementation arms and usual care and conditional vs unconditional delivery. KCCQ-CSS change from baseline to 90 days was analyzed using linear mixed-effects models with fixed effects for treatment group, time (categorical), and treatment-by-time interaction, plus a random intercept for each participant, using all available data. Adjusted models included the same covariates as the HF hospitalization or ED visit for HF model. Results were presented as least-squares means with 95% CIs and between-group differences. The odds of achiev-

Table 2. Feasibility, Adherence, and Implementation Outcomes by Food Supplementation Group

Outcome	Medically tailored meals	Fresh produce	P value ^a
Delivery completion			
Scheduled deliveries, No.	555	558	NA
Completed deliveries, No. (%)	500 (90.1)	542 (97.1)	NA
Weekly held deliveries, No. (%)	55 (9.9)	16 (2.9)	NA
Reasons for weekly paused delivery			
Conditionality related, No. (%)	16 (29.1)	14 (87.5)	NA
Not conditionality related, No. (%)	39 (70.9)	2 (12.5)	NA
Participant opted out	39	1	NA
Held due to hospitalization	0	1	NA
Participant-reported adherence, d/wk ^b			
Overall, mean (SD)	4.7 (2.4)	5.5 (2.3)	.001
1-mo Follow-up	4.9 (2.2)	5.3 (2.4)	.14
2-mo Follow-up	4.3 (2.6)	5.8 (2.0)	.002
3-mo Follow-up	4.9 (2.3)	5.2 (2.4)	.60
Implementation outcomes, mean (SD) ^c			
Acceptability	3.62 (1.07)	4.16 (0.74)	.005
Appropriateness	3.81 (0.92)	4.07 (0.69)	.12
Feasibility	4.13 (0.81)	4.29 (0.55)	.26
Net Promoter Score ^d			
Score, mean (SD)	7.27 (3.14)	8.62 (2.46)	.02
Promotional, No. (%)	17 (37.8)	34 (72.3)	
Passive, No. (%)	14 (31.1)	6 (12.8)	.004
Detractive, No. (%)	14 (31.1)	7 (14.9)	

Abbreviation: NA, not applicable.

^a P values represent comparisons across randomization groups using the Wilcoxon rank sum test.^b Self-reported adherence was assessed by asking participants to indicate the number of days in the past week (of 7) that they consumed at least 1 meal (breakfast, lunch, or dinner) originating from the fresh produce or medically tailored meal box. Adherence was assessed on a monthly basis.^c Implementation outcome measures assessed participants' perceptions of the acceptability, appropriateness, and feasibility of the intervention. Each domain consists of 4 items rated on a 5-point Likert scale.^d The Net Promoter Score is a customer experience and loyalty metric that evaluates how likely participants are to recommend the program to others. It categorizes respondents' contribution into promotional (score 9-10), passive (score 7-8), and detractive (scores 0-6).

ing an improvement of 10 points in KCCQ-CSS from baseline to 90 days were analyzed using multivariable logistic regression. Process-of-care outcome measures were compared between groups with Wilcoxon rank sum tests. Additional exploratory analyses assessed whether implementation outcomes and intervention effects differed by baseline food security status. All statistical tests were 2-sided with $\alpha = .05$. Analyses were performed using R version 4.3.1 (R Foundation for Statistical Computing).

Results

Of 222 eligible patients hospitalized with HF who were invited to a screening visit, 150 participants were enrolled and randomized (Figure 1). No screened participant was excluded for existing meal delivery program participation. Participants were allocated 1:1:1 to medically tailored meals ($n = 50$), fresh produce ($n = 50$), or usual care with dietary counseling only ($n = 50$). Patients randomized to food supplementation underwent secondary 1:1 randomization to delivery conditional ($n = 50$) or unconditional ($n = 50$) on health care engagement. During 90-day follow-up, 2 participants were withdrawn due to clinical deterioration, 1 died, and 6 were lost to follow-up. All 150 randomized participants were included in the intention-to-treat clinical outcome analyses.

The median (IQR) age was 59.5 (52.0-66.0) years; 91 (60.7%) were male and 59 (39.3%) were female; 2 participants (1.3%) were

Asian, 63 (42.0%) were Black, 80 (53.3%) were White, and 5 (3.3%) other; 49 (32.7%) were Hispanic. Fifty-seven participants (38.0%) were uninsured or had county-sponsored coverage. At baseline, the median (IQR) left ventricular ejection fraction was 35% (25-54), and the median (IQR) KCCQ-CSS was 56.6 (36.8-72.9). Overall, 99 (66.0%) had either food insecurity or nutrient insecurity. Baseline characteristics were generally similar across the 3 randomization groups (Table 1).

Intervention Delivery, Feasibility, Acceptability, and Costs

Among participants randomized to food supplementation, delivery rates were high, with 93.6% of deliveries successfully completed (Table 2). Among 71 paused deliveries, 30 (42.3%) were due to conditionality-related nonengagement with medical care. Most delivery pauses that were not conditionality related were in the medically tailored meals group (39 of 41 [95.1%]). Mean (SD) self-reported weekly food supplement consumption was higher in the fresh produce (vs medically tailored meals) group (5.5 [2.3] vs 4.7 [2.4] days per week; $P = .001$) (Table 2).

Participants rated the fresh produce boxes (vs medically tailored meals) as more acceptable (mean [SD] rating, 4.16 [0.74] vs 3.62 [1.07], $P = .005$), with similar feasibility and appropriateness. Mean (SD) participant satisfaction, measured by the Net Promoter Score, was significantly higher for fresh produce boxes (vs medically tailored meals; 8.62 [2.46] vs 7.27 [3.14], $P = .02$). The average 7-day cost was \$105 for fresh produce boxes and \$122 for medically tailored meals (14 meals/7

Table 3. Effects of Food Supplementation vs Usual Care^a

Outcome	Any food supplementation vs usual care	Medically tailored meals vs usual care	Fresh produce vs usual care
Primary outcome			
HF readmissions or ED visits through 90 d, aRR (95% CI)	1.09 (0.49 to 2.43)	1.29 (0.53 to 3.16)	0.95 (0.37 to 2.43)
P value	.83	.57	.92
Secondary outcomes			
Win-ratio hierarchical composite (95% CI)	1.21 (1.14 to 1.29)	NA	NA
P value	<.001		
Adjusted between-group difference in KCCQ-CSS at 90 d, β (95% CI)	7.30 (−0.91 to 15.51)	7.10 (−2.94 to 17.14)	7.37 (−2.22 to 16.96)
P value	.08	.16	.13
KCCQ-CSS improvement ≥ 10 points at 90 d, aOR (95% CI)	2.09 (1.01 to 4.31)	1.58 (0.69 to 3.59)	2.80 (1.16 to 6.72)
P value	.046	.28	.02

Abbreviations: aOR, adjusted odds ratio; aRR, adjusted rate ratio; ED, emergency department; HF, heart failure; KCCQ-CSS, Kansas City Cardiomyopathy Questionnaire Clinical Summary Score; NA, not applicable.

^a All models adjusted for study site, age, sex, ejection fraction, and baseline estimated glomerular filtration rate. β represents the adjusted between-group difference in predicted KCCQ-CSS at 90 days from linear mixed-effects models additionally adjusted for baseline score, time, and treatment-by-time interaction. Win-ratio hierarchical composite: (1) all-cause death, (2) total hospitalizations or ED visits for HF, (3) ≥ 10 -point KCCQ-CSS improvement at 90 days.

days, approximately \$8 per meal), including home delivery. Total 90-day food costs were \$1260 per participant for fresh produce and \$1464 per participant for medically tailored meals.

Association of Food Supplementation With Clinical Outcomes

During 90-day follow-up, there were 32 HF hospitalizations or HF ED visits (27 of 150 [18.0%] experienced events) (Table 3). The rate of HF hospitalizations or ED visits did not differ significantly between pooled food supplementation arms (23 events among 100 participants) and usual care (9 events among 50 participants) (adjusted rate ratio [aRR], 1.09; 95% CI, 0.49–2.43; $P = .83$) and between each food supplementation group vs usual care (Table 3). For the hierarchical composite outcome, food supplementation (vs usual care) was associated with a favorable win ratio (1.21; 95% CI, 1.14–1.29; $P < .001$) (Figure 2). There were greater wins vs losses for a ≥ 10 -point improvement in KCCQ-CSS with food supplementation (1510/5000 [30.2%] wins vs 962/5000 [19.3%] losses).

Longitudinal changes in KCCQ-CSS for food supplementation vs usual care and across individual food supplementation groups vs usual care are shown in eFigure 1 in Supplement 2. There was no significant between-group difference in KCCQ-CSS between food supplementation and usual care at 90 days (β , 7.30; 95% CI, −0.91 to 15.51; $P = .08$) (Table 3). Similar findings were observed between each food supplementation group vs usual care. When a threshold change in KCCQ-CSS was evaluated, food supplementation was associated with greater odds of achieving a ≥ 10 -point improvement in KCCQ-CSS compared with usual care (adjusted odds ratio [aOR], 2.09; 95% CI, 1.01–4.31; $P = .046$). This benefit was seen in the fresh produce vs usual care (aOR, 2.80; 95% CI, 1.16–6.72; $P = .02$), with no significant difference for medically tailored meals vs usual care (aOR, 1.58; 95% CI, 0.69–3.59; $P = .28$).

Association of Conditional vs Unconditional Delivery With Clinical Outcomes

Baseline characteristics were similar across groups conditioned on health care engagement (eTable 5 in Supplement 2).

Among participants randomized to conditional vs unconditional food interventions, there were no differences in the 90-day rate of HF hospitalizations or ED visits for HF (aRR, 1.04; 95% CI, 0.43–2.48; $P = .93$) (eTable 6 in Supplement 2). The hierarchical outcome significantly favored conditional vs unconditional delivery (win ratio, 1.37; 95% CI, 1.25–1.50; $P < .001$). Participants receiving conditional delivery had significantly higher KCCQ-CSS at 90 days than those receiving unconditional delivery (β , 9.60; 95% CI, 0.48–18.71; $P = .04$) (eFigure 1 in Supplement 2). Conditional delivery was associated with significantly higher KCCQ-CSS at 90 days than usual care (β , 11.45; 95% CI, 1.53–21.38; $P = .02$) without significant difference in unconditional delivery vs usual care (β , 2.01; 95% CI, −7.75 to 11.76; $P = .69$). There were no significant differences between conditional and unconditional delivery groups in clinic visit attendance (83% vs 89%; $P = .07$) or medication fill rates (94% vs 88%; $P = .18$) (eTable 7 in Supplement 2).

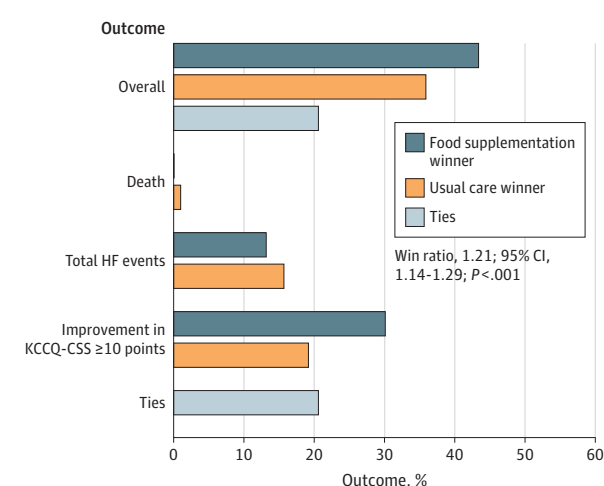
Association of Treatment With Outcomes Across Food Security Status at Baseline

Implementation outcomes did not differ significantly between food insecure and food secure participants (eTable 8 in Supplement 2). There was no significant heterogeneity in effects for HF hospitalizations or ED visits by food security status (eTable 9 in Supplement 2).

Discussion

In this randomized clinical trial of 150 patients recently hospitalized for HF, food supplementation for 90 days was feasible and well accepted, and fresh produce boxes demonstrated superior acceptability, with higher satisfaction scores and greater participant adherence compared with medically tailored meals. Food supplementation did not reduce the primary clinical outcome of readmission or ED visits for HF compared with usual care. However, exploratory outcomes identified a difference in odds of achieving a ≥ 10 -point improvement in KCCQ-CSS with food supplementation vs usual care and a significant differ-

Figure 2. Bar Chart Showing Win Ratio Outcome and Components



Visual representation of the hierarchical composite outcome analyzed using win-ratio methodology for pooled food supplementation groups vs usual care. The hierarchical composite prioritized components as (1) all-cause death, (2) total heart failure (HF) hospitalizations or emergency department visits, and (3) clinically meaningful improvement in health status (≥ 10 -point increase in Kansas City Cardiomyopathy Questionnaire Clinical Summary Score [KCCQ-CSS]). The win ratio is calculated as total wins divided by total losses.

ence in the hierarchical composite between arms. These findings establish the feasibility of food supplementation among HF patients in the vulnerable postdischarge period and provide insights for the design of future food supplementation trials.

Prior food supplementation evaluations in patients with HF have been limited to observational studies or small trials with heterogeneous results. The Geriatric Out of Hospital Randomized Meal Trial in Heart Failure (GOURMET-HF) pilot trial¹⁴ (N = 66) found no QOL improvement with 4 weeks of DASH-compliant meals vs usual care following HF hospitalization. In contrast, the Nutritional Intervention Program in Malnourished Patients Admitted for Heart Failure (PICNIC)¹⁵ trial among patients with malnourishment and HF demonstrated reductions in mortality and readmission with 6 months of intensive nutritional support. In the Kaiser Permanente Evaluation of Medically Tailored Meals in Adults With Medical Conditions at High Readmission Risk (KP-NOURISH) trial²² (N = 2000 with cardio-kidney-metabolic conditions), food supplementation showed no effect on all-cause hospitalization. However, in the HF subgroup (n = 641) a 47% reduction was observed in HF-specific readmissions. The FOOD-HF trial extends existing literature as a prospectively designed trial examining food supplementation after HF hospitalization and comparing 2 different food supplementation strategies. Like GOURMET-HF, our study targeted patients with HF with high burden of disease severity in the vulnerable postdischarge period where potential benefits of food supplementation are greatest. The cohort in our trial was enriched for patients with food insecurity (66%), though we also enrolled participants with food and nutrient security to evaluate broad population feasibility. Despite targeting a

socioeconomically vulnerable patient population with high disease severity, the trial achieved high intervention fidelity, high participant satisfaction, and favorable ratings across acceptability, appropriateness, and feasibility measures.

The FOOD-HF intervention prioritized vegetables, lean proteins, whole grains, and sodium restriction (<2000 mg daily), adapting DASH diet principles for HF management. This approach builds on recent trials demonstrating cardiometabolic benefits of dietary interventions. Specifically, the Groceries for Black Residents of Boston to Stop Hypertension Among Adults Without Treated Hypertension (GoFresh) trial²³ showed 12 weeks of home-delivered DASH groceries reduced systolic blood pressure by 3.4 mm Hg in Black adults with hypertension, although the effects were not sustained postintervention. Further, the Coronary Artery Risk Development in Young Adults-Salt Sensitivity of Blood Pressure (CARDIA-SSBP) trial²⁴ demonstrated that a highly controlled 1-week low-sodium diet reduced blood pressure by 8 mm Hg in middle-aged adults. Our study builds upon these dietary intervention trials but addresses a distinct population and clinical context. Unlike prior trials targeting low-risk outpatients, FOOD-HF enrolled HF patients in the immediate postdischarge period. Our analyses showed no significant reduction in HF hospitalizations or ED visits but were underpowered. It is also possible that the effects of dietary interventions may require longer intervention. The improvement in the hierarchical, win-ratio endpoint provides important insights. There were greater odds of achieving clinically meaningful improvement of 10 or more points on the KCCQ-CSS. However, food supplementation did not show a significant improvement in continuous KCCQ-CSS at 90 days.

Our pilot findings can inform future larger trials powered for clinical end points. The superior acceptability of fresh produce boxes likely reflects participants' preference for preparing meals from provided ingredients. Intervention costs are modest and align with existing coverage through insurance plans that already contract with commercial vendors.²⁵ Leveraging this infrastructure enables pragmatic trial designs^{22,26} whereby insurance covers food costs while research funds support outcome assessments. This allows for simultaneous evaluation of clinical effectiveness and real-world implementation, facilitating rapid translation if effective. Future trials should incorporate a community-engaged design with culturally adapted recipes, multilingual nutrition education, evaluate longer duration interventions (6 to 12 months), and consider food insecurity as an enrichment criterion or stratification factor.

To our knowledge, FOOD-HF is the first study to test whether food supplementation can incentivize downstream health care engagement, building on behavioral economics principles that demonstrate incentives can effectively modify health behaviors.^{27,28} Our approach was consistent with recent trials leveraging loss aversion, the principle that avoiding losses motivates behavior more than equivalent gains.^{29,30} Participants receiving conditional food supplementation suggested greater likelihood to have improvement in KCCQ-CSS, whereas those receiving unconditional provision showed none

compared to diet counseling alone. However, conditional delivery was not significantly associated with health care process outcomes, with both groups demonstrating high clinic visit attendance and medication fill rates.

Limitations

This study has limitations. First, this was a 2-site pilot trial and was underpowered to detect clinically meaningful differences in HF events. Second, given heterogeneity of nutritional content, frequency, and duration, our results may not extend to other food supplementation approaches. Third, while food insecurity was assessed at baseline, formal nutritional status was not, and food insecurity may not correlate with nutritional deficits. Fourth, the open-label design may have introduced bias, particularly for patient-reported outcomes like KCCQ-CSS. Fifth, we relied on validated quantitative measures rather than qualitative methods for implementation outcomes. Sixth, we did not

assess local food environment, limiting our understanding of the context in which participants selected meals beyond supplemented foods. Seventh, the 3-month intervention duration may have been insufficient to observe the full clinical impact of sustained nutritional support.

Conclusions

This pilot randomized clinical trial demonstrated the feasibility of food supplementation in patients recently hospitalized with HF, with fresh produce supplementation demonstrating greater acceptability, satisfaction, and adherence compared to medically tailored meals. Food supplementation did not reduce HF hospitalization or HF ED visits. These pilot findings can inform larger, multicenter trials investigating food supplementation in the postdischarge setting for patients with HF.

ARTICLE INFORMATION

Accepted for Publication: February 3, 2026.

Published Online: April 8, 2026.

doi:10.1001/jamacardio.2026.0435

Author Contributions: Drs Pandey and Segar had full access to all of the data in the study and take responsibility for the integrity of the data and the accuracy of the data analysis.

Concept and design: Pandey, Keshvani, Coellar, Peterson.

Acquisition, analysis, or interpretation of data: All authors.

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Critical review of the manuscript for important intellectual content: Keshvani, Coellar, Segar, Rizvi, Peterson.

Statistical analysis: Pandey, Segar.

Obtained funding: Pandey, Peterson.

Administrative, technical, or material support:

Pandey, Keshvani, Coellar, Jain, Bustillo-Rubio, Rizvi.

Supervision: Pandey, Coellar, Rizvi, Peterson.

Conflict of Interest Disclosures: Dr Pandey reported research support from the National Institute of Health, American Heart Association, Applied Therapeutics, Roche, Ultrametrics, Gilead Sciences, Bayer, and AstraZeneca as well as honoraria as an advisor/consultant for Tricog Health, Eli Lilly, Rivus, Cytokinetics, Roche Diagnostics, Axon Therapies, Medtronic, Edward Lifesciences, Science37, Novo Nordisk, Bayer, Medical AI, AstraZeneca, Baylor Scott & White Research Institute, Boehringer Ingelheim, iRhythm Technologies, Tourmaline Bio, Merck, Sarfex Pharmaceuticals, Semler Scientific, Ultrametrics, Encarda, Tenax Pharma, Alnylam, Abbott, Kilele Health, Anumana, Acorai, Novartis, and Antlia Biosciences. Dr Keshvani reported personal fees from Science37 during the conduct of the study. Dr Segar reported nonfinancial support from Pfizer and Merck and personal fees from Otsuka and Idorsia outside the submitted work. Dr Peterson reported grants from Amgen and personal fees for study executive committee participation from Janssen and Novo Nordisk outside the submitted work. No other disclosures were reported.

Funding/Support: This work was funded by the American Heart Association (Food as Medicine grant RPGFIM1198188 to Drs Pandey and Peterson).

Role of the Funder/Sponsor: The funder had no role in the design and conduct of the study; collection, management, analysis, and interpretation of the data; preparation, review, or approval of the manuscript; and decision to submit the manuscript for publication.

Data Sharing Statement: See Supplement 3.

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